

Mixed Reality as a Tool to Enhance Mathematics Education in Elementary School

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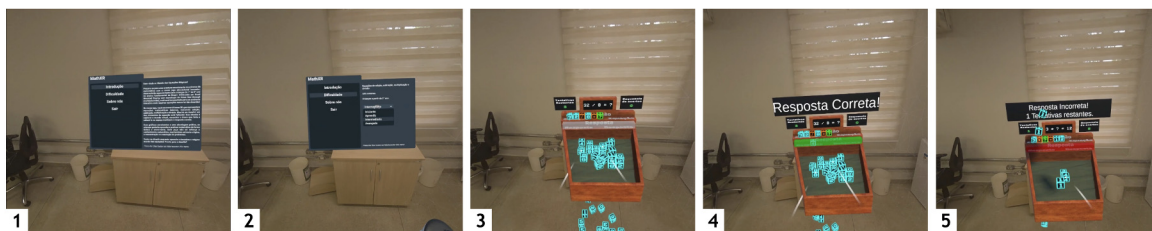


Figure 1: Screen captures of the developed game running on the Meta Quest 3 HMD: 1) Game menu; 2) Game instructions; 3) Example quiz; 4) Correct answer example; 5) Wrong answer example.

ABSTRACT

This paper investigates the use of mixed reality technology to support mathematics instruction at the elementary level. An MR-based educational game focused on the four basic operations was developed and tested with a group of elementary school teachers, who provided feedback on its usability and potential for classroom integration. The results suggest that MR-based learning not only supports a deeper conceptual understanding of mathematics but also addresses a variety of learning preferences, creating a more inclusive and engaging educational environment. The study highlights the necessity for ongoing teacher training and resource development to fully harness the potential of MR in elementary school mathematics education.

Index Terms: Augmented Reality, quiz, game, headset.

1 INTRODUCTION

Teaching mathematics to young learners lays a foundation for future academic progress, particularly as technology and computational thinking become more common in modern education. When introducing new skills, it helps to present concepts in a way that encourages absorption and retention. However, many educators face limited student engagement and a shortage of effective teaching resources.

Mixed reality (MR), a blend of physical and virtual environments, offers one avenue for reimagining elementary-level mathematics instruction. By providing interactive, visually appealing experiences, MR-based tools can make abstract operations (e.g., addition, subtraction, multiplication, and division) more accessible and

captivating for students. The present study explores how these tools can influence student engagement and comprehension, focusing on an MR-powered educational game aimed at strengthening the four fundamental operations.

The main objective is to evaluate whether an MR approach can enhance both students' understanding of mathematical concepts and their motivation to learn. Through prototype testing and discussions with K–5 educators, this work examines the perceived benefits, challenges, and long-term implementation potential of MR in early math education. Initial findings point to the possibility of more inclusive learning experiences, as teachers identify the technology's capacity for addressing the diverse needs of students and offering solutions to enduring obstacles—such as limited motivation and lack of resources—in the field of mathematics education.

2 LITERATURE REVIEW AND BACKGROUND

The integration of educational technologies, particularly MR, into the contemporary educational environment has shown significant potential to transform teaching and learning. According to Tori, Hounsell, and Kirner [8], virtual and augmented reality offer an effective way to combine virtual and physical elements, enriching the educational experience through interactivity and immersion.

Significant research on MR in K–12 education has demonstrated various advantages and potential pitfalls. Radu [5] provides a comparative review of how augmented reality (AR) can positively influence learning outcomes, including (i) increased content understanding, (ii) improved long-term memory retention, (iii) heightened motivation, and (iv) better collaboration. These findings underscore that AR can be a powerful asset for mathematics education. However, the same author warns of challenges such as excessive attention focus, usability difficulties, and ineffective integration in the classroom.

Recent studies have also highlighted the broader impact of digital games in education. Lee [4] compared the effects of AR mobile games and printed games on student engagement and attitudes in foreign language learning, finding that both mediums can sustain high levels of engagement and positive attitudes. Drakatos et al. [3]

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explored the use of online gaming in engineering education, emphasizing the motivational potential of games to enhance learning outcomes and student engagement. Young et al. [9] reviewed trends in serious gaming for education, noting that while evidence for the impact of games on academic achievement is still emerging, games have shown promise in areas such as language learning and physical education.

In addition, Borborema and Mauricio [1] emphasize that AR in Education 4.0 not only enhances interactivity and engagement in learning environments, but also highlights the need for teachers to develop digital competencies to effectively integrate AR as a pedagogical tool.

Bujak et al. [2] offer a psychological framework that analyzes AR-based learning across three main dimensions: physical, cognitive, and contextual. Their analysis highlights the importance of the spatiotemporal alignment of abstract content (e.g., numbers or geometric shapes) with the real environment, thereby alleviating cognitive load. Furthermore, physical manipulation of learning objects in AR can strengthen memory retention through embodied cognition. In a math classroom, this means that arithmetic operations, fractions, or geometry can become more intuitive and concrete when students see digital content anchored to physical objects or their own gestures.

Although the research by Radu [5] and Bujak et al. [2] suggests that AR can be highly beneficial in the educational process, the teacher's perspective is crucial for the successful use of technology in the classroom. Radu [6] recommends using a participatory design approach, involving the teacher in the co-creation of AR prototypes. The interviewed teachers considered certain areas of mathematics, such as word problems, nonstandard measurement, fractions, and place value, as particularly challenging to teach with conventional methods. By aligning the AR prototypes with these areas, the teachers felt more optimistic about addressing the conceptual challenges students face. However, the study also notes that educators often require both technical and pedagogical support to integrate AR, emphasizing the need for iterative testing and teacher feedback cycles.

These studies assert that there is synergy between MR in K-12 mathematics, combining virtual objects with tangible interactions. As stated in the studies by [2, 5], the integration of embodied learning principles, where students physically engage with concepts, can deepen mathematical understanding and retention.

The present work aims to continue this line of investigation by examining educators' opinions on the impact of a mixed-reality educational game for teaching mathematical operations, focusing on their perceptions of student motivation and performance in elementary education math classes, using the mixed-reality educational game as a pedagogical resource.

3 PROPOSED APPLICATION

In this paper, we propose and evaluate an educational game that uses mixed reality to teach the four arithmetic operations (MR-based math quiz) to elementary education math classes. This solution draws on the literature above, particularly focusing on strategies that harness AR's capacity for visualizing numbers and operations in physical space, VR's immersion for motivation, and game-based interaction to reinforce core math concepts.

We developed an MR game for the Meta Quest 3 headset (Fig. 1). The game integrated real-world surfaces, the passthrough feature, and interactions with digitally rendered math puzzles. We chose vibrant colors for the 3D blocks to capture children's attention and simplify object discrimination. The manipulation of objects is possible through the controllers.

Development began with creating interactive math equations, a process that incorporated 3D visual elements to represent math problems, as described by Schiavoni [7], who explores the logic

of space in virtual reality to facilitate conceptual understanding. These elements were then integrated into a quiz environment that tests and reinforces students' math knowledge at various difficulty levels. The development of the game was carried out in five stages, as detailed below:

1. **Generating Math Equations:** A C# script was coded to automatically generate basic math equations where one of the elements is unknown (e.g., " $4 + ? = 9$ ") (Fig. 2). The equations vary according to difficulty level:

- **Level 1:** Addition and subtraction of tens;
- **Level 2:** Addition, subtraction, multiplication, and division of tens;
- **Level 3:** All basic operations up to hundreds; and
- **Level 4:** All basic operations up to thousands.

This approach was inspired by the idea of offering progressive scaffolding mentioned in [6].

2. **Instantiating Equations in the 3D World:** The system instantiates 3D numeric blocks in the virtual environment, connecting the content with the real environment (Fig. 3). This concept was suggested by Bujak et al. [2] on how to leverage spatiotemporal alignment to reduce cognitive load.



Figure 2: Example of a basic math equations generated with an unknown term.

3. **Reading User Responses:** We applied colliders as triggers to detect which blocks the students place in specific spots, converting these block positions into numerical text variables (Fig. 4). This feature promotes active and embodied reasoning in problem-solving [5].

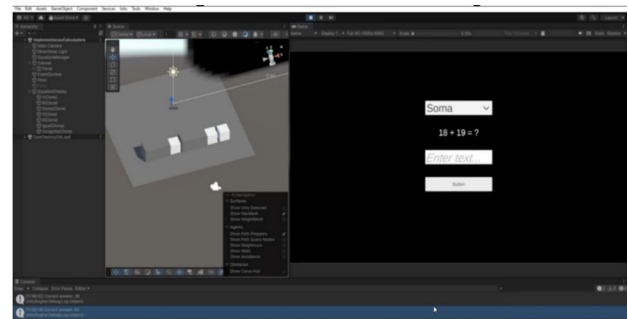


Figure 3: Block elements being arranged over a reference surface.

4. **Creating and Positioning the 3D Board:** A 3D board organizes equations, triggers, and numeric blocks (Fig. 5). By

positioning the board on real-world surfaces, we enable exploration from multiple perspectives, an approach that fosters deeper spatial understanding, as identified by Bujak et al. [2].

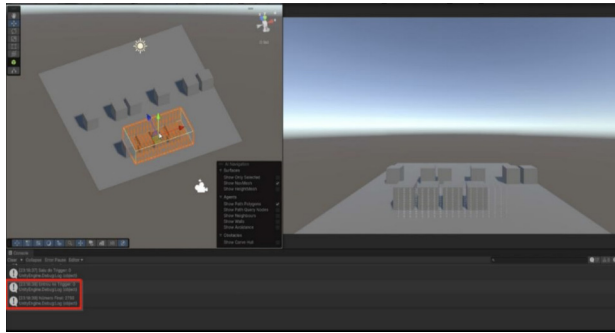


Figure 4: Orange box showing a collider responsible for detecting blocks being manipulated.

5. **Managing the Game Stages:** A script was coded to coordinate the gameflow: (i) environment setup; (ii) equation display; and (iii) feedback loops (Fig. 6).



Figure 5: 3D board representing the work area of the game. The blocks should be manipulated (grabbed and placed) inside this board.

To support immediate and context-relevant feedback [5], correct answers lead to a new equation, while incorrect answers encourage multiple attempts and scaffolds.

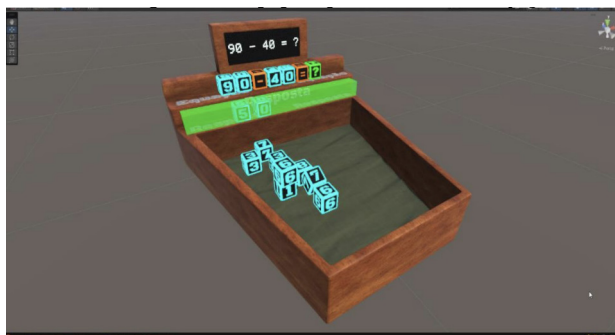


Figure 6: In this example, the green area represents the place where the answer blocks should be placed for instantaneous feedback.

4 RESULTS VALIDATION

To validate the MR-based math quiz and examine educators' opinions on its impact for teaching mathematical operations, an experiment was conducted as follows. Ten teachers from Gabriela Mistral school were invited to participate in the experiment. They primarily teach students in Grades 1 through 5 and agreed to participate in the validation. The teachers who participated in the experiment are between 24 and 58 years old. During the testing sessions, each teacher interacted with the AR application for about 5 to 7 minutes, exploring the environment, positioning the 3D board, and solving a series of basic equations (Fig. 7). After the interaction, the participants completed a questionnaire with subjective questions and were interviewed to further elaborate on their impressions of the game as a pedagogical resource. In the interview, the participating educators emphasized that they believe the game is more suitable for students in the 4th and 5th grades, as the game's level of abstraction and interactive content align well with the math curriculum typically covered in these grades.

An online questionnaire was administered to teachers to gather insights about their familiarity with immersive technologies, their views on how these tools might improve mathematics education, and how they perceive potential challenges and benefits of using AR for teaching. Below is a summary of the most significant patterns and observations:

Most teachers reported being "a little familiar" with AR and VR technologies, while two indicated having no familiarity at all. None identified themselves as "very familiar." This suggests that, although there is some awareness of immersive tools, in-depth knowledge remains uncommon among the participants, thus indicating a need for training or professional development before large-scale classroom adoption.

A majority believe that immersive technologies could enhance math education at the elementary level. Several teachers selected "Yes," expressing optimism about MR's capacity to make math concepts more interactive and appealing. Three teachers responded with uncertainty ("I am not sure"), highlighting that some educators may need more concrete examples or success stories to be fully convinced.

Asked about the obstacles their students face, the teachers commonly cited:

- Lack of student interest and motivation
- Little use of educational technologies
- Specific cognitive difficulties of students
- Inadequate or outdated teaching methods
- Lack of support and educational resources at home

These responses underscore the complexity of teaching math in elementary school: in addition to cognitive challenges, teachers see an urgent need for pedagogical innovation and technological integration.

All respondents deemed the development of new educational technologies "very important," signaling a strong desire for novel strategies to bolster math learning. This enthusiastic consensus suggests an openness within the teaching community to adopt modern, learner-centered solutions.

Most participants rated an MR game for addition, subtraction, multiplication, and division as either "very effective" or "effective." This finding aligns with their beliefs about improving engagement—teachers often stated that immersive technologies could overcome motivational barriers by offering more interactive and visually stimulating experiences.

When asked about the value of MR research for educational technology and pedagogical innovation, the vast majority again

rated it as “relevant” or “very effective” in terms of potential to inform teacher training or to integrate MR more broadly in classrooms. Some teachers explicitly mentioned that the project could promote the integration of MR in elementary schools and contribute to teacher professional development programs.

A clear majority indicated they would “whenever possible” use AR in their teaching. A smaller group selected “sometimes.” In terms of ease of use, most reported it as “neutral” or “easy,” although a few found it “difficult.” Interestingly, regarding the Meta Quest headset usability, teachers tended to rate it “easy” or “neutral,” signaling that while the headset is manageable, it may still require some support or guidance.



Figure 7: Teacher assessing the augmented reality game.

One teacher recommended providing a step-by-step guide or orientation for new users, underscoring that user-friendly guidance materials would help reduce any initial confusion. This aligns with the overall feedback regarding the need for practical information on how to embed the technology smoothly into daily classroom routines.

The data suggest that teachers:

- Recognize the potential of MR to increase student engagement and improve learning outcomes in math,
- Feel that inadequate use of technology is a major contributor to current math-learning difficulties,
- Strongly advocate for developing innovative tools, particularly for essential operations and to address motivational gaps,
- Desire some form of training, guidelines, or professional development resources to ease adoption.

Teachers displayed cautious enthusiasm. While most see a clear benefit in applying immersive technologies, the shift to MR must be accompanied by adequate training in teacher Continuing Professional Development and straightforward instructional resources to ensure its successful integration into the elementary math curriculum. This feedback confirms that although there is genuine promise in using MR for education, practical concerns, such as ease of use, teacher familiarity, and alignment with curricular goals, remain key determinants of widespread implementation.

5 CONCLUSION

The project demonstrates how MR-based math quizzes can facilitate fundamental operations learning by merging dynamic visualizations, embodied interactions, and real-world references. Our findings echo earlier research that AR can provide substantial motivation, deeper conceptual understanding, and a greater sense of presence compared to traditional classroom methods. However, challenges remain, including hardware costs, teacher training in continuing professional development, and ensuring a balanced focus on both “fun” and “function” in AR-based learning tools. Future work will expand on hand-tracking interactions, incorporate more robust gamification, and investigate how such a solution might adapt to diverse student populations to deliver inclusive and effective AR-based mathematics instruction.

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